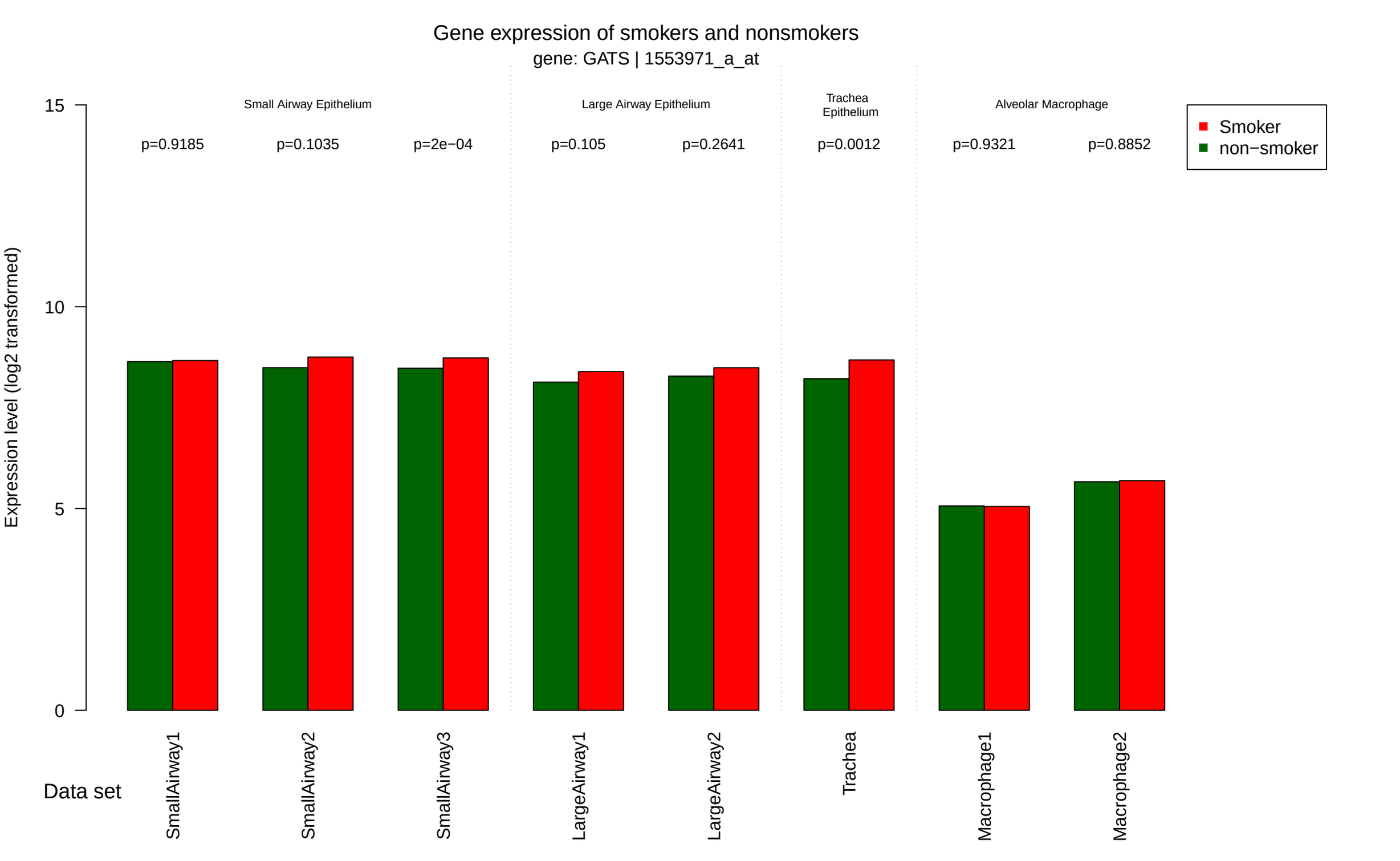
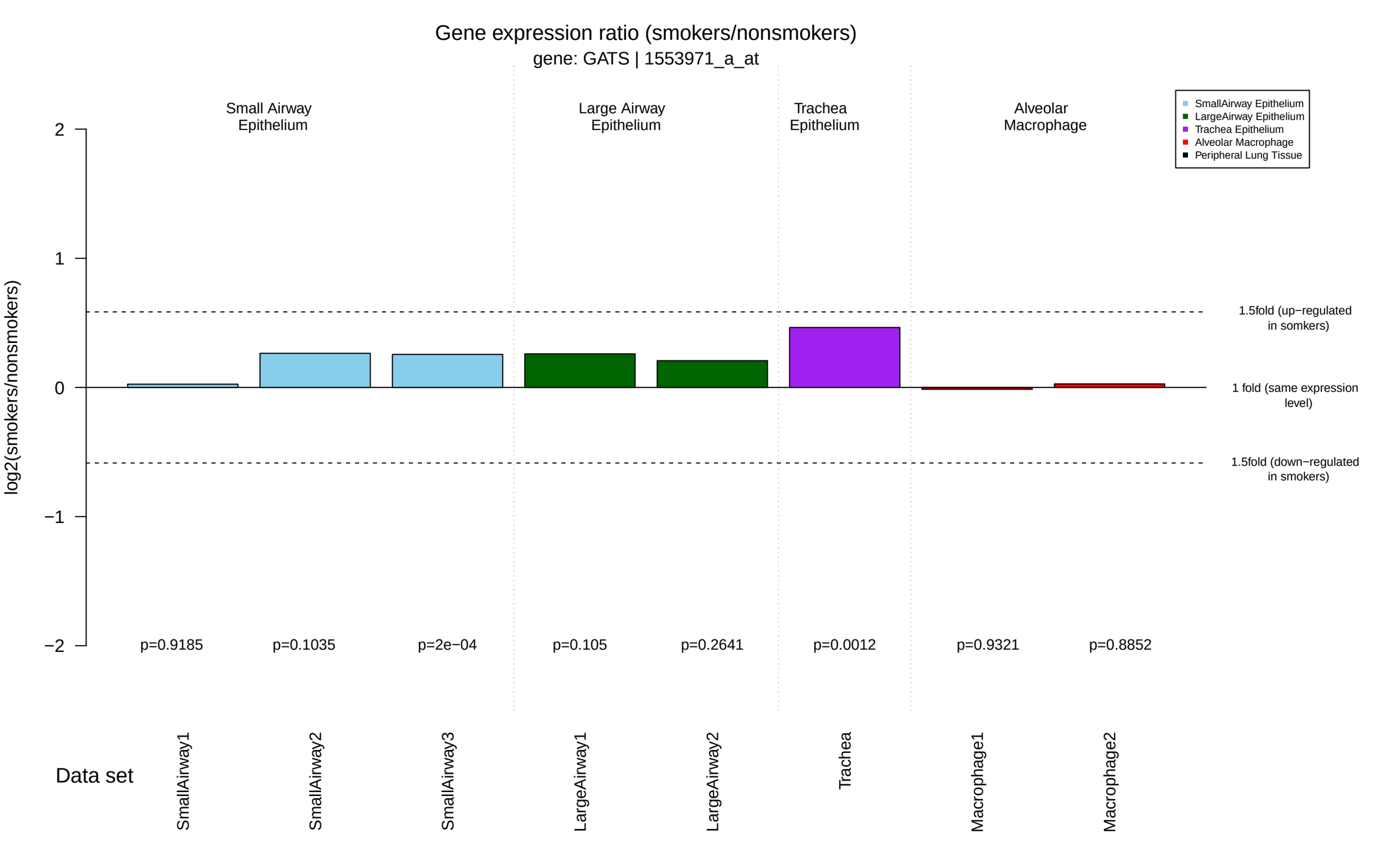
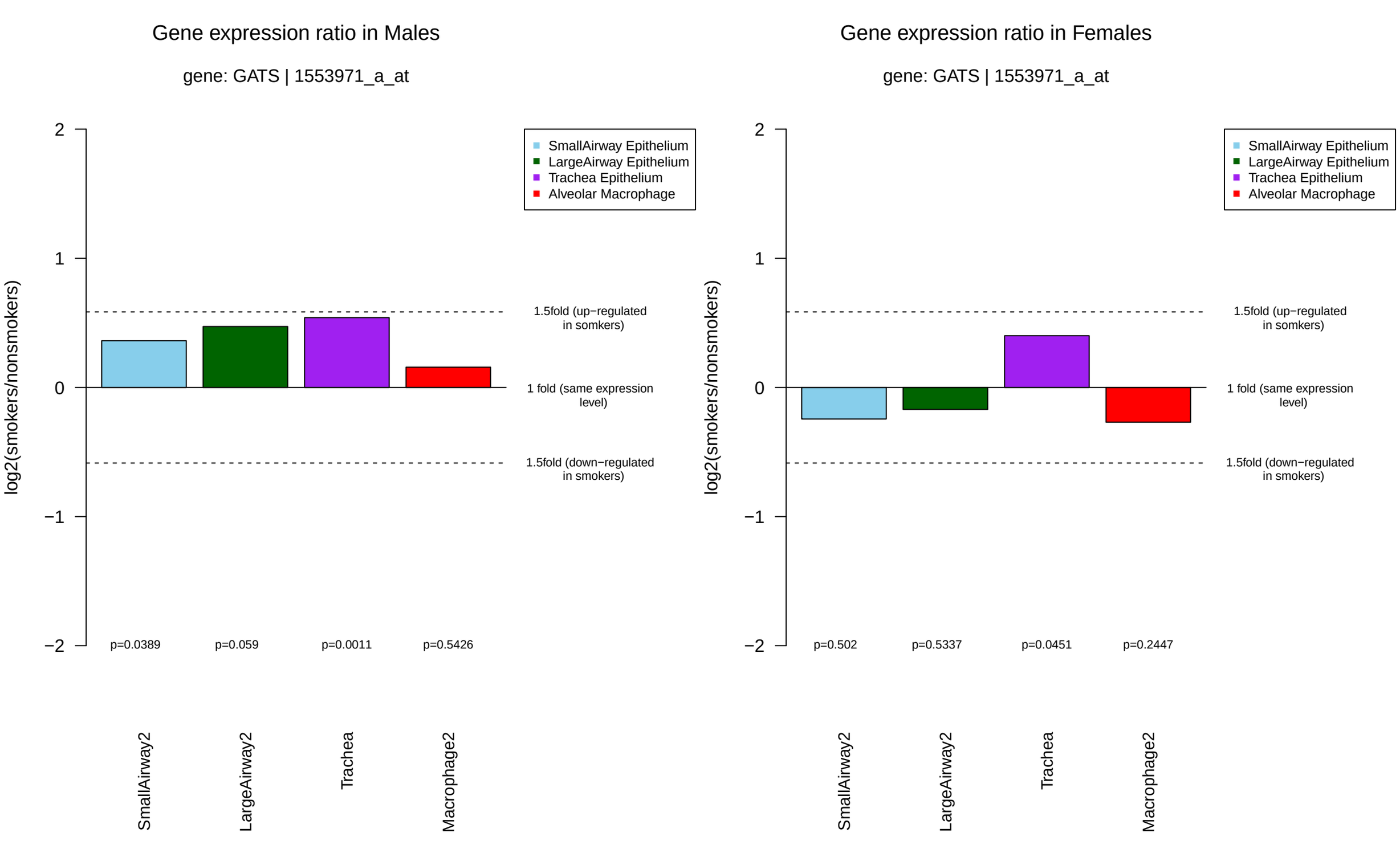
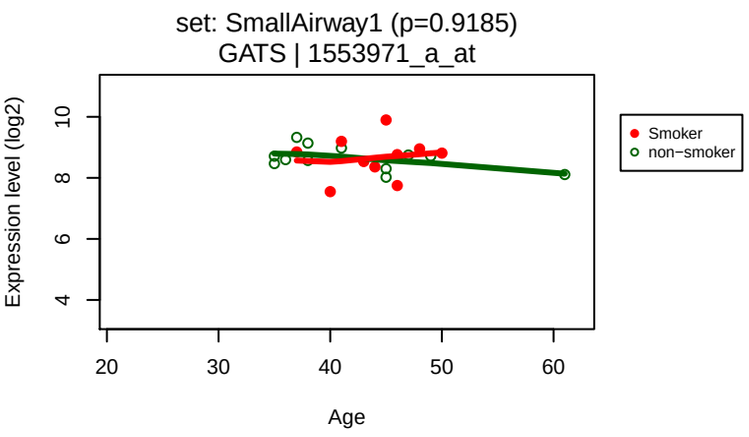
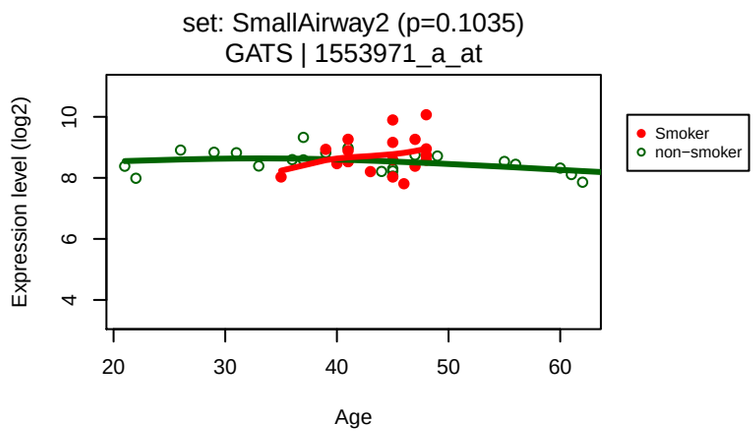
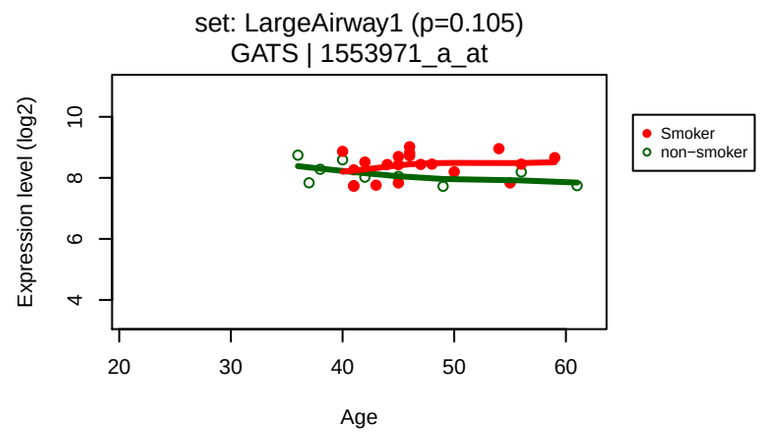

Gene expression of smokers and nonsmokers
gene: GATS | 1553971_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9185
p=0.1035
p=2e−04
p=0.105
p=0.2641
p=0.0012
p=0.9321
p=0.8852
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

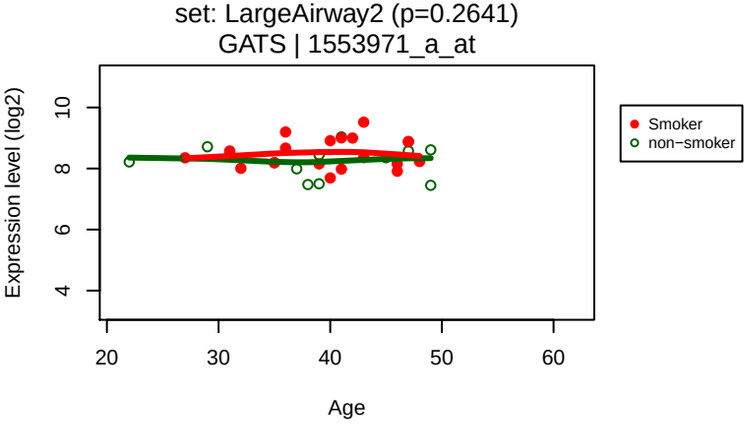

Gene expression ratio (smokers/nonsmokers)
gene: GATS | 1553971_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9185
p=0.1035
p=2e−04
p=0.105
p=0.2641
p=0.0012
p=0.9321
p=0.8852
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

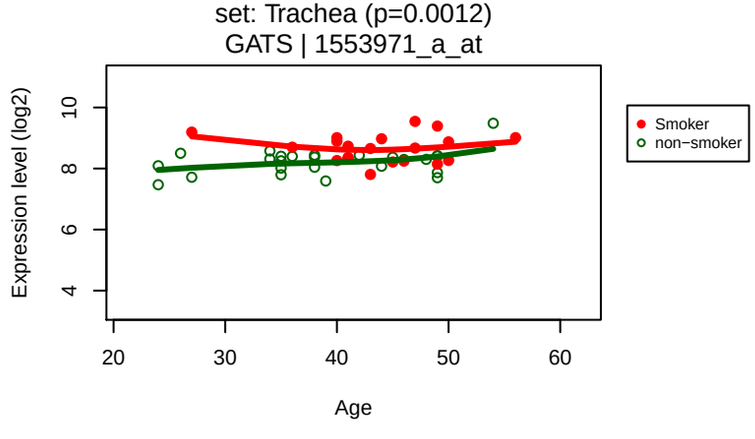

Gene expression ratio in Males
gene: GATS | 1553971_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0389
p=0.059
p=0.0011
p=0.5426
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GATS | 1553971_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.502
p=0.5337
p=0.0451
p=0.2447
SmallAirway2
LargeAirway2
Trachea
Macrophage2

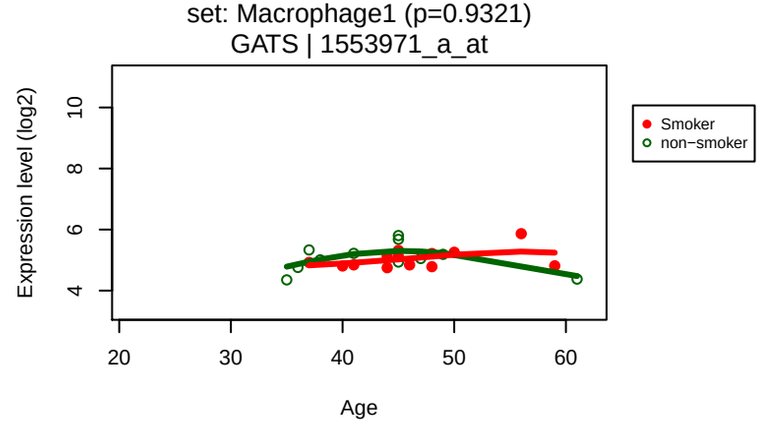

set: SmallAirway1 (p=0.9185)
GATS | 1553971_a_at
Expression level (log2)
Age
Smoker
non−smoker

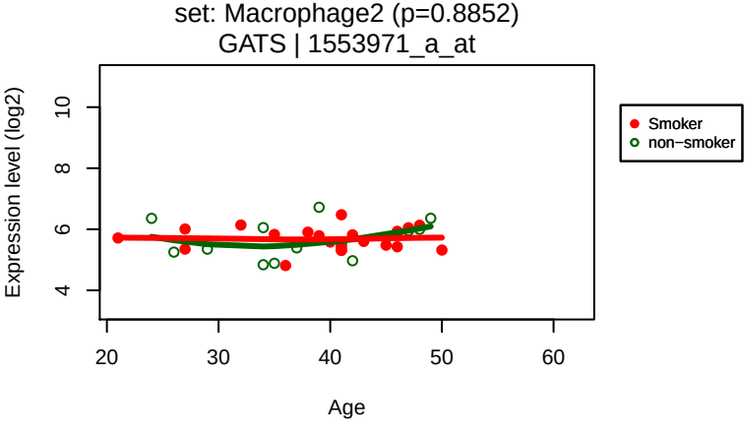

set: SmallAirway2 (p=0.1035)
GATS | 1553971_a_at
Expression level (log2)
Age
Smoker
non−smoker

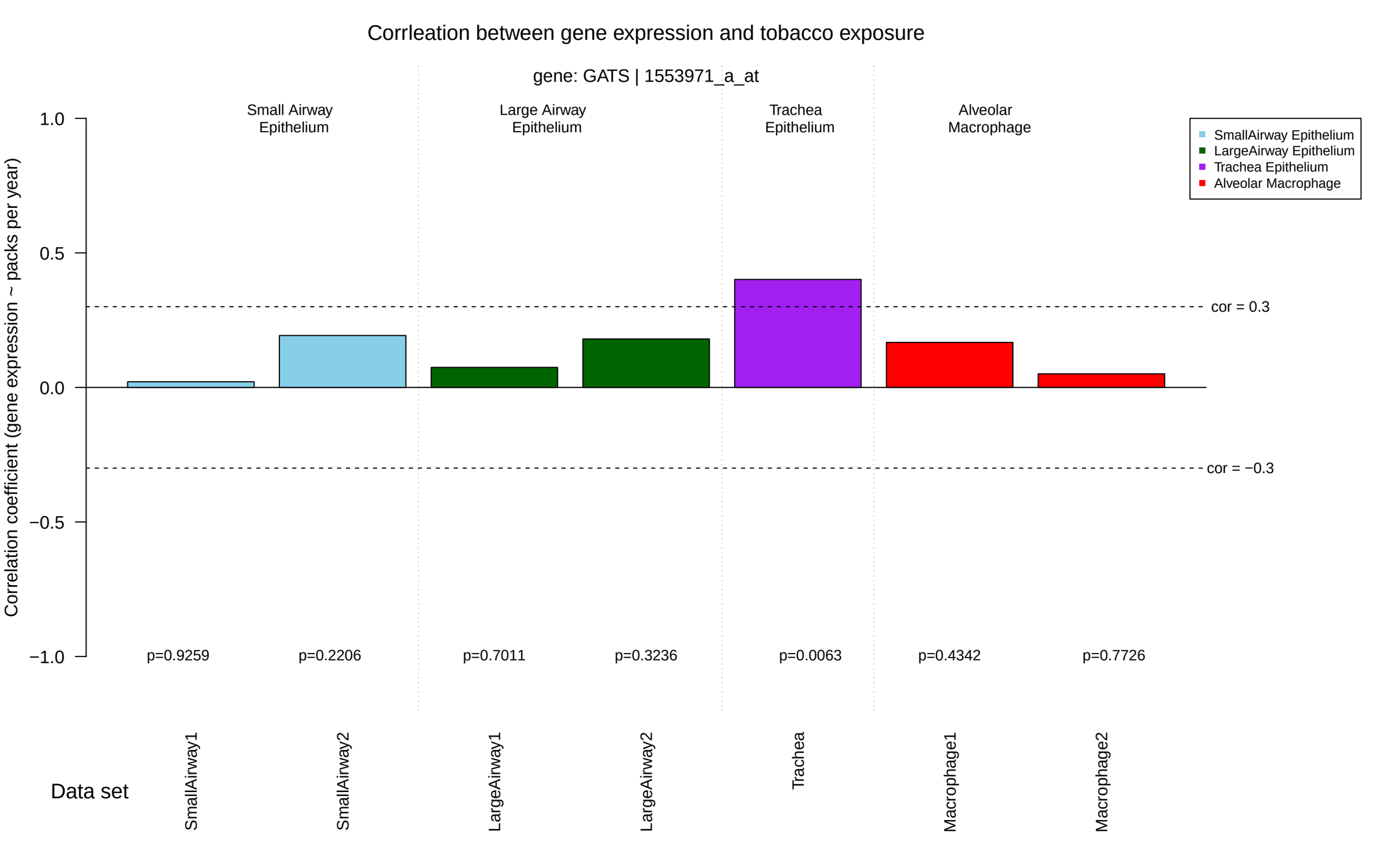

set: LargeAirway1 (p=0.105)
GATS | 1553971_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2641)
GATS | 1553971_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0012)
GATS | 1553971_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9321)
GATS | 1553971_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8852)
GATS | 1553971_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GATS | 1553971_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9259
p=0.2206
p=0.7011
p=0.3236
p=0.0063
p=0.4342
p=0.7726
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2